# Read the axes before you read the line

GUIDED NOTES / use with the projected diagram Name _____

## A SAY THE WORD

### Signed area / ES: área con signo / PT: área com sinal

Height × width = displacement. Area below zero is negative displacement.

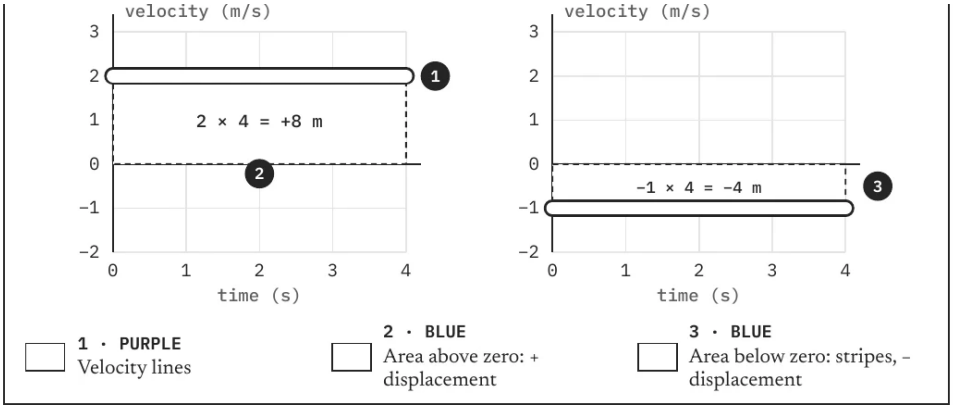
Also on your note page: Velocity–time graph, Constant velocity, Components.

## B READ THE RELATIONSHIP

### Signed area on a velocity–time graph gives displacement.

Displacement = signed area under a velocity–time graph

## C TRACE THE ORIGINAL FIGURE




Use the numbered key. Trace one feature. Tell your partner what it means.

## D FOLLOW THE WORKED EXAMPLE

### Velocity is +3 m/s for 2 s. Find the displacement and name its direction.

**Given** v = +3 m/s t = 2 s

**Rule** $\Delta x = v \times t$

**Work** $\Delta x = (3 \text{ m/s})(2 \text{ s})$

**Answer** +6 m 6 m to the right

## E SAVE FOR THE FINAL MEETING

A velocity–time graph shows whether XJ's speed is holding steady. Splitting its velocity arrow into parts shows how much is aimed at Earth and how much carries it sideways.

# Read the physics: Signed area

SUPPORTED READING / read one part, then point to the evidence Name _____

## 01 PICTURE IT

Read the vertical axis first. A flat line on a velocity–time graph means steady velocity. A flat line above zero means the object is moving.

## 02 HOW IT WORKS

Shade the region between velocity and the time axis. For steady velocity, multiply velocity by time. Add areas above zero; subtract the sizes of areas below zero. For distance, add all the sizes.

## 03 WHAT TO CHECK

A diagonal velocity has two components. For an angle measured from +x, use cosine for the x part and sine for the y part.

## 04 TRY THE EXAMPLE FROM YOUR NOTES

Velocity is +3 m/s for 2 s. Find the displacement and name its direction. +6 m 6 m to the right.

## 05 CONNECT TO XJ

A velocity–time graph shows whether XJ's speed is holding steady. Splitting its velocity arrow into parts shows how much is aimed at Earth and how much carries it sideways.

SAY IT BEFORE YOU WRITE

Which shaded part subtracts from displacement? The area is ___ because it is ___ zero.

# Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

## Velocity–time graph

ES: gráfica velocidad-tiempo / PT: gráfico velocidade-tempo

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain velocity–time graph to a friend?<br>In my words, this means ___. | Which axis helps you tell whether the cart is moving left or right?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| How would you show three seconds of steady motion at −2 m/s?<br>My example shows ___. | A student says a flat line at +2 means the cart is parked. What does that height mean?<br>I would change ___ because ___. |

## Constant velocity

ES: velocidad constante / PT: velocidade constante

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain constant velocity to a friend?<br>In my words, this means ___. | Photographs are one second apart. What pattern would you look for?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| Draw three positions for a cart moving 2 m right each second.<br>My example shows ___. | A bike circles a track at a steady speed. Is its velocity constant?<br>I would change ___ because ___. |

# Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

## Signed area

ES: área con signo / PT: área com sinal

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain signed area to a friend?<br>In my words, this means ___. | What do you do with the area below the time axis when you find displacement?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| Sketch a rectangle at −2 m/s lasting 3 s. What displacement does it show?<br>My example shows ___. | Someone labels the area −2 m/s. Which graph reading did they use?<br>I would change ___ because ___. |

## Components

ES: componentes / PT: componentes

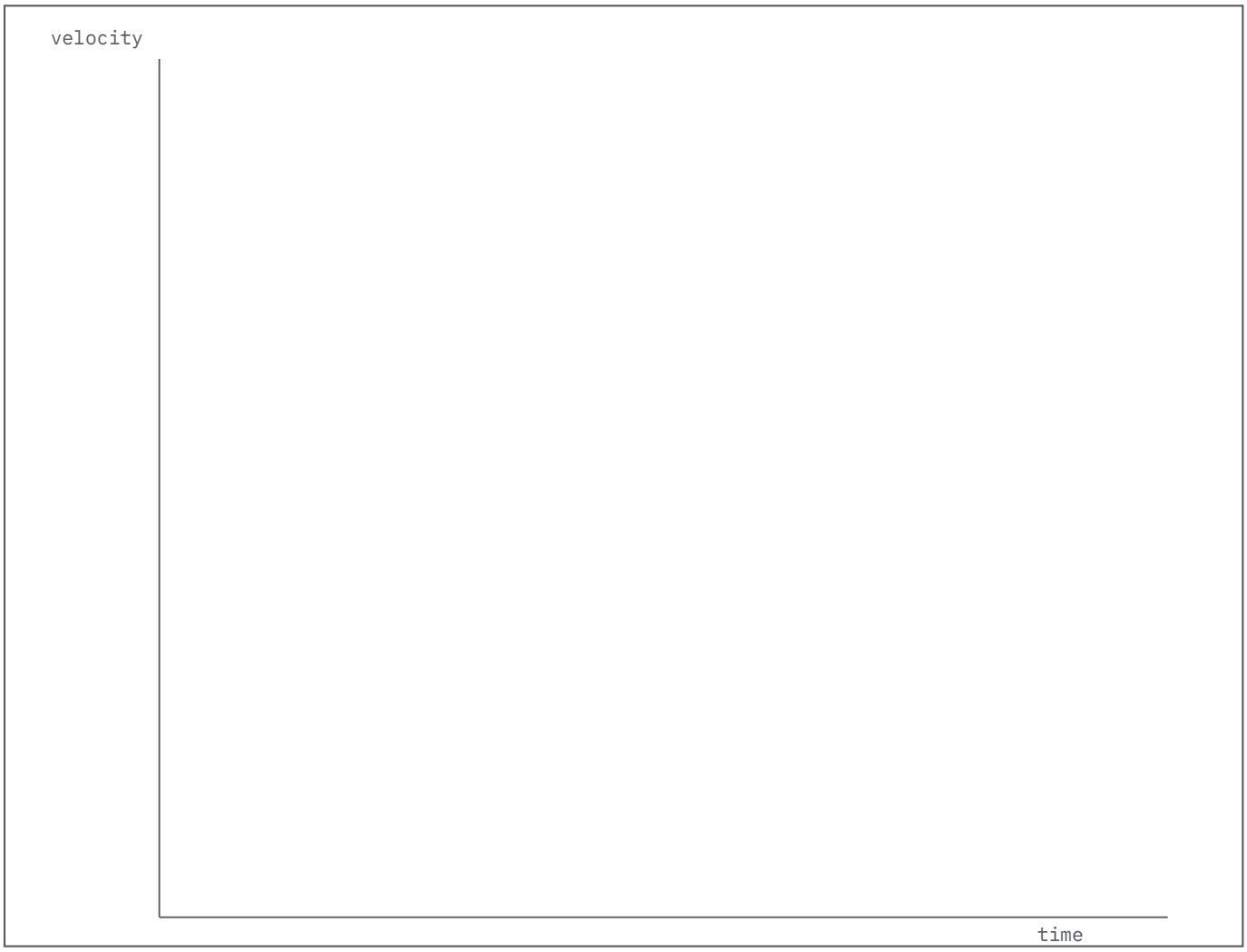
| Meaning in my words | How can you tell? |
|---|---|
| What are the components of an arrow?<br>In my words, this means ___. | If east and north are positive, which component is negative for a southeast trip?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| What two arrows could you use to build a displacement of 3 m east and 4 m north?<br>My example shows ___. | A classmate counts the diagonal and both parts as three separate trips. What is counted twice?<br>I would change ___ because ___. |

# Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

## Draw and shade opposite-side rectangles. Put m/s on the vertical and s on the horizontal axis.

Labels you can use: above · below · area.




**Tell a partner**

The area is ___ because it is ___ zero.

# Where does the camera finish?

04

**SECTION E** FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

**Our goal:** Use signed graph areas to find displacement.

**CHOOSE ONE JOB TOGETHER**

**TRACKING ENGINEER** ☐

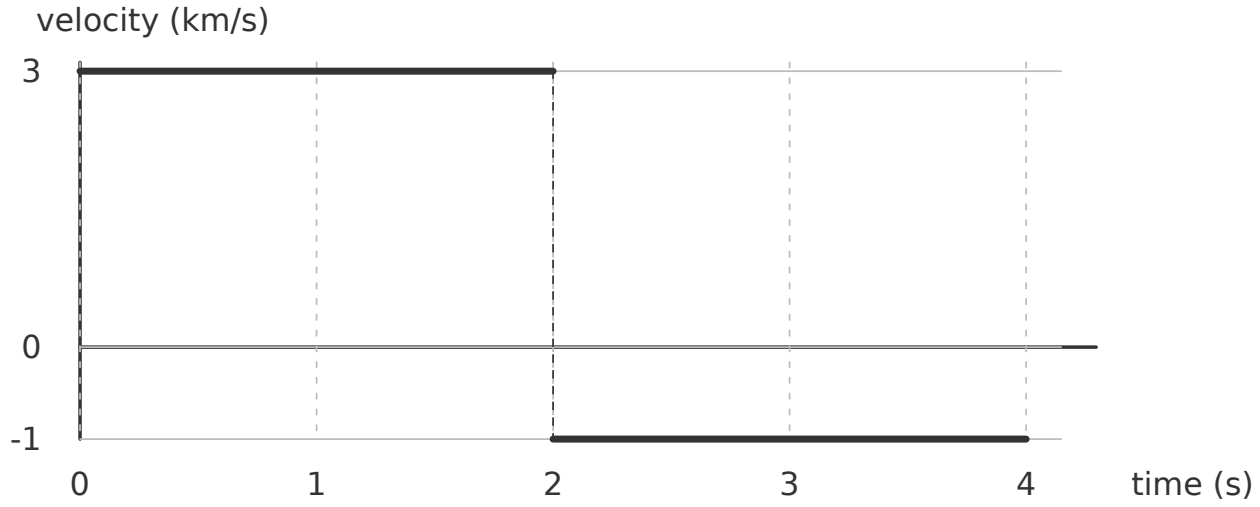
Shade both parts of the velocity graph. Radio the camera's final change in position, including direction.

**REPLAY PRODUCER** ☐

Mark the outward and return shots. Report total distance and net movement to the editor.

**Word help:** area / return / displacement

A moving camera follows a model at **+3 km/s** for **2 s**, then **−1 km/s** for **2 s**. The editor needs the end location and the total footage distance.




## 01 OUR FIRST CALL

**Where does the camera finish relative to its start? How far did it travel?**

The camera travels ___ but ends ___.

## 02 OUR REVISED CALL

**Wait for the reveal. Keep your first call.**

New clue: _____

Check the new caption. Which number belongs on the location graphic? Explain.

Now we would say ___ because ___.

**Switch speakers.** Did you count the return trip with the right sign?

| ☐ GREEN | ☐ RED | ☐ YELLOW | ☐ BLUE | ☐ PURPLE | ☐ ORANGE | ☐ PENCIL |
|---|---|---|---|---|---|---|
| speeding up | slowing down | steady / balanced | position / displacement | velocity | forces | other marks |